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Explains BIOS, UEFI, long mode transitions, `bootloader_api` mechanisms, ELF loading, paging setup, and memory identity mappings.

## Memory Management

Extremely detailed exploration of physical memory maps, virtual address spaces, paging structures, MMU behavior, TLB design, and planned memory allocators.

## CPU Architecture

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## VGA Subsystem

Full write-up of text mode memory, hardware scroll behavior, color registers, character cell behavior, memory race hazards, volatile memory, and synchronization.

## Driver System

Planned driver model, dispatch tables, interrupt-driven drivers, polling drivers, keyboard driver, timer driver, block device concepts, and future plug-and-play interface.

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